

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Start-up and Commissioning Procedure
Document Number	EA-BFPCL-GPMC-GGPL-FEED- OPS-PRC-001
Document Revision	R02
Document Status	Issued for Review
Originator	Olarinde Ige
Security Classification	Restricted
Issue Date	20-May-22

Start-up and Commissioning Procedure

R02	20.05.22	Issued for Review	O.Ige	M.Ekpunobi	K.Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	13.05.22	First Issue for Clients Review
R02	20.05.22	Comments Implemented
A01		

Review Code & Status (to be filled by Client)

☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- OPS-PRC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

Table of Contents

1.	INTRODUCTION	5
1.1.	Background Information	5
1.2.	Purpose	6
1.3.	Scope	6
2.	REGULATIONS, CODES AND STANDARDS	7
2.1.	Abbreviations	7
3.	GENERAL SAFETY REQUIREMENTS	9
3.1.	Risk Assessment	9
3.2.	PTW System	9
4.	ROLES AND RESPONSIBILITIES	10
5.	COMMUNICATION STRATEGY	11
6.	COMMISSIONING PROCESS AND DEFINITIONS	12
6.1.	Process Description	12
6.2.	Preparatory Activities	12
6.3.	Mechanical Completion	12
6.3.1.	General preservation and cleanliness	12
6.3.2.	Electrical	12
6.3.3.	Instrumentation and Telecommunications Equipment	12
6.3.4.	Mechanical	12
6.3.5.	Flange Tightening/Torquing/Tensioning	13
6.3.6.	Joint Integrity Management	13
6.4.	Mechanical Completion Check List	14
7.	PRE-COMMISSIONING CHECKS	15
7.1.	Piping & valves	15
8.	COMMISSIONING STRATEGY	17
8.1.	Steps for Commissioning and Start-up:	17
9.	RFSU/PSSR/PSUA	18
9.1.	PSUA / PSSR	18
10.	PROJECT HAND-OVER	19
10.1.	Required Documents	19
10.2.	System Handover Documentation	19
10.3.	Operations P&ID Check List	20

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opubene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

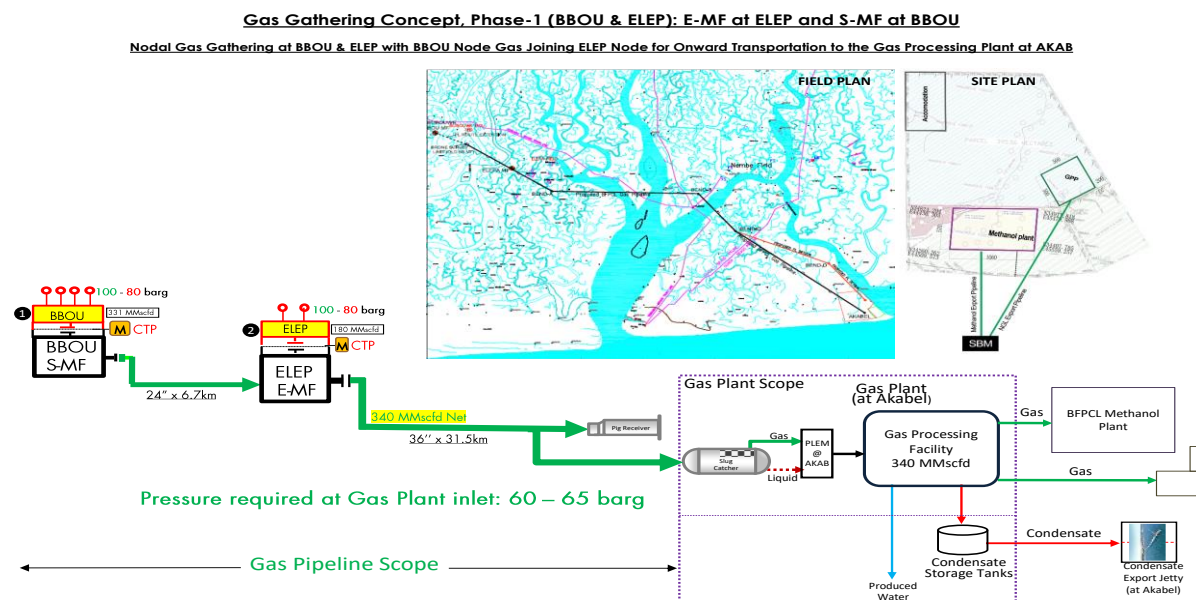


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this Start-up and Commissioning Procedure is to provide a structured guideline to be followed during the commissioning of the gas pipelines and associated infrastructure to ensure safety of personnel and assets, with due regard to the environment.

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Scope

The Facilities to be covered under the project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

2. Regulations, Codes And Standards

Discipline	Document Number	Title
Piping	API 1104	Welding of Pipeline and Related Facilities
Piping	ASME B31.3	Process Piping
Piping	ASME B31.8	Gas Transmission and Distribution Piping Systems
Piping	AWS D1.1	Structural Welding
Piping	ASNT SNT-TC-1A	American Society for Non-Destructive Testing, recommended practice
Piping	ASTM A694	Carbon and Alloy Steel Forgings for Pipe Flanges. Fittings, Valves and Parts for High Pressure Transmission Service.
Piping	ASME B 16.34	Valves – Flanged Threaded and Welded Ends
Piping	API 598	valve Inspection and test

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold

BFPCL	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Sales & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License
OSD	Operational Shut Down
PAS	Process Automation System
PC	Pressure Controller
PCV	Pressure Control Valve
PTW	Permit to Work
SPDC	Shell Petroleum Development Company of Nigeria

3. General Safety Requirements

The start-up and commissioning activities for the BFPCL gas gathering pipelines and field manifolds shall be guided by strict compliance to the following safety requirements;

Safety is the responsibility of every employee and all employees shall ensure that their work- place is safe and free of obstructions that could result into accident. Safe working practices shall be implemented and adhered to at all time. Particular attention will be given to worksite safety rules, and compliance to any safety systems and procedures provided.

Personal Protective Equipment (PPE) will be issued to all personnel. The minimum PPE of hard hat, safety boots, full length coveralls, gloves and eye protection shall be worn as a minimum at all times during the commissioning activities.

Worksite Barriers shall be erected in order to ensure that unauthorized personnel do not enter the worksite, and safety warning signs shall be conspicuously displayed to warn personnel of site dangers during the commissioning activities.

3.1. Risk Assessment

HSE Risk Assessments shall be carried out for the commissioning activities. The process will identify hazards and the degree of risk that they pose. Control measures shall be indicated for the identified risks (safety equipment, work procedures, isolation methods, training and communication) which are required to minimize the risks to an acceptable level. Assessments shall be carried out prior to commencement of Commissioning activities.

The Risk Assessment procedure requires completion of the schedule of tasks/operations for the activity sheet. This sheet is used to give a description of the task and highlight whether it involves hazards to persons, plant or asset. If the task involves hazards, the risk assessment analysis sheet must be completed.

3.2. PTW System

The permit-to-work (PTW) system should ensure that authorised and competent people have thought about foreseeable risks and those risks are avoided by using suitable precautions. Those carrying out the job should think about and understand what they are doing to carry out their work safely and take the necessary precautions for which they have been trained and made responsible.

Contained within the PTW system are other precautions which may need to be taken into consideration - e.g., process or electrical isolation, or access barriers - and these will need to be identified in task risk assessments (Job Hazards Analysis) before any work is undertaken.

4. Roles and Responsibilities

Pre-commissioning and commissioning activities shall be carried out by the Project team with full participation of the Operations and Maintenance team, who will be an integral part of the commissioning team.

However, when the facilities are ready for introduction of hydrocarbons after a successful Pre-Start up Audit (PSUA) or Pre-Start up Safety Review (PSSR), the Operations team shall be responsible for introduction of hydrocarbons, with the assistance of the Vendors.

5. Communication Strategy

During the Commissioning and Start-Up of the BFPCL gas gathering pipelines and associated infrastructure, all communication will be via radio/Mobile phone and Email among all parties involved.

Prior to commencing Commissioning and Start-up, A PSSR in addition to a Start-Up on Paper (SUOP) shall be conducted to ensure the Start-up procedure are fully understood, the requirements in event of an emergency are known, the roles and responsibilities are identified, the necessary interfaces are known and the communication protocol for the Start-Up phase is fully understood.

6. Commissioning Process and Definitions

6.1. Process Description

The major activities involved in commissioning include the following; Mechanical Completion, preparatory, pre-commissioning (pre-start-up audit, leak test, functionality test, instrument loop test), commissioning and acceptance performance test.

6.2. Preparatory Activities

These include all preparations identified by the Project or the Operations Readiness & Assurance (OR&A) Rep which are required for effective execution of commissioning, including all necessary vendor specialist training for future process operating and maintenance personnel during the plant commissioning. They include organizing budget, planning, scheduling, and ensuring the availability of relevant updated design, engineering, vendor, and construction documents.

6.3. Mechanical Completion

6.3.1. General preservation and cleanliness

A system of planned preservation activities (preservation plan), combining standard nature of preservation activities and specific vendor requirements, shall be established by the Project team.

6.3.2. Electrical

The Inspection and Testing of Electrical equipment and installed circuits shall be carried out in accordance with the approved Inspection and Test Record Sheets and approved supporting Project Procedures. Verification that such Inspections have been carried out satisfactorily shall be recorded by Sub-contractor QA/QC & BFPCL QA/QC on the applicable Inspection and Test Record form.

6.3.3. Instrumentation and Telecommunications Equipment

The Inspection and testing of Instrumentation equipment and installed circuits shall be carried out in accordance with the Inspection and Test Sheets and supporting Procedures. Verification that such Inspections have been carried out satisfactorily shall be recorded by Sub-contractor QA/QC & BFPCL QA/QC on the applicable Inspection and Test Record form.

All field installed instruments external to Vendor skids shall be fully calibrated by the equipment supplier or approved delegate and should be supplied to site with a calibration certificate.

6.3.4. Mechanical

Generally, any Mechanical equipment within a packaged skid would have been inspected and tested during the Factory Acceptance Test (FAT) and will not require detailed Inspection and test at site. Only selected Inspection and Test sheets will be applied on key equipment with a specific

purpose or alternately.

Vendors site Inspection procedures shall be utilized as certification to verify the equipment is installed correctly to specification prior to Commissioning.

Flange alignments shall be checked and confirmed to be stress free. The alignment shall be performed following detailed project procedures.

The bolting on main process connections shall be removed if deemed necessary to check for pipe spring prior to testing. Re alignment checks shall be carried out if any remedial work is required.

Where no Commissioning Strainers exist, all upstream piping and vessels shall be checked following hydro testing and flushing to confirm the absence of any debris or other contaminants that could lead to machinery damage or premature wear.

6.3.5. Flange Tightening/Torquing/Tensioning

Stud bolts will have sizes checked and recorded to ensure that it is in accordance with the Project specifications. As a general guideline the following criteria may be complied with: -

- For hand tightened (using torque bar) & Hydraulically Torqued joints minimum 2 threads (or 6mm) extend out sided the fully tightened nut.
- For Hydraulic Tensioned joints – minimum 1.5xDiameter of bolt shall extend outside the fully tightened nut.

No flogging of bolts or other uncontrolled methods of joint tightening or break out shall be permitted.

Flanges shall be made up and tightened in accordance with approved methods detailed within the approved Project specifications and procedures & details provided below, for each bolting material type, size, coating, and type of lubricant applied.

6.3.6. Joint Integrity Management

Verification of the correct joint make up and tightening method shall be recorded on the appropriate reinstatement Inspection and Testing Sheets during the Mechanical Completion phase and these shall be recorded and verified by the OR&A Rep.

Where project contract term requires, special flange management and integrity test for selected hydrocarbon system & high-risk pipe work will be performed as per specialist contractor work procedures. The timing of such special integrity management shall be towards the end of the commissioning activities.

Such special requirement for hydrocarbon & high-risk pipe work systems, the specialist contractor shall implement a Flange Integrity Management program in order to tag, track, record and verify

that every flanged joint on each critical system has been correctly assembled, tightened, torqued or tensioned, successfully leak tested inclusive of records of any remedial work found necessary.

6.4. Mechanical Completion Check List

In general, Mechanical Completion shall include the activities listed below. More detailed activities are included in the checklists below;

- Piping systems shall be checked against designs, cleaned, tested for pressure ratings and tightness, dried, inerted, and restored to tight condition in readiness for operation.
- All equipment shall be installed, assembled, inspected, calibrated, adjusted, cold-aligned, tested, lubricated, and cleaned as necessary to be ready for operation.
- Elements such as strainers, temporary strainers, and filters shall be installed, cleaned or replaced so that a clean element is in place for commissioning.
- All electrical systems shall be installed, energized, and tested for continuity, insulation resistance, ground fault, and operational functions.
- All instrumentation shall be installed, loop checked, set to proper set points, and calibrated.
- Instrumentation loop checks to the PLC, field control panels and other control systems shall be completed.
- Probes, coupons, samplers, and other consumables shall be installed.
- Utility and safety systems shall be commissioned and demonstrated to be operational.
- All painting, coating, and insulation work shall be completed.
- All changes and modifications shall be incorporated.
- All functional tests and Site Acceptance Test of individual systems shall be performed.
- All work shall be verified to be in accordance with the related Specifications, Drawings (PTT and vendor), codes, and standards.
- Deficiencies identified during Mechanical Completion activities shall be remedied prior to the commencement of Commissioning.
- Mechanical Completion documentation shall be completed and signed.

7. Pre-Commissioning Checks

7.1. Piping & valves

Piping shall be properly installed according to Project P&IDs and isometrics, including in line valves, instruments, piping specialty items, supports, coating, insulation, and tracing are complete. Stress supports shall be inspected to confirm they are installed in accordance with the design drawings and validated against the Stress Isometric load requirements.

Line identification tags, stencilled lines and signage/colour banding are completed prior to Mechanical Completion and shall be checked to ensure it conforms to project specification.

Checks shall be performed to ensure that all temporary supports, piping connections, blinds/spades/spacers, sea fastening, bracing and stops have all been removed and confirmed as part of Line Walk checks with the OR&A Rep.

Checks shall be performed to ensure Start Up strainers or filters have been installed, as required, prior to MC and are clearly marked and a Master Temporaries register maintained.

Check shall be performed to ensure valve interlocks, blanks & Spec Blinds etc. are completed prior to MC.

Installation of all piping specialty items such as chemical injection quill fittings, sampling point fittings etc. are verified as part of re-instatement Inspection Reports.

All piping work scope up to the stage of final reinstatement shall be the responsibility of the appropriate construction sub-contractor who shall keep all certification on file and verified through BFPCL QA/QC process.

Each piping pressure test will have its own unique test pack number assigned, be allocated to its sub-system, and recorded as part of the Project Handover documentation.

For Pipeline cleanliness the Project philosophy is that when piping systems are presented as 'Ready for MC handover', they are as clean as possible within a construction environment. The piping cleanliness activity prior to MC handover shall be achieved either thru;

- a. Water flushing prior to hydrotesting & draining/drying completely
- b. Chemical cleaning & passivation for selected piping
- c. Oil flushing for selected piping, and
- d. For very large diameter pipes manual wire brushing & air jetting.

Results of flushing shall be recorded with in flushing Inspection and Test reports, which shall be part of piping test pack.

Hydro Testing shall be carried out in accordance with approved procedures and recorded on the Inspection Test Record Sheet and shall include details of sections tested, location of blinds, valves and inline equipment removed, vent and drain locations, chart recorded records & calibration certification of test equipment.

Valves shall be pressure tested at the manufacturer's works to the applicable codes and standards.

Also checks shall be conducted to verify that all the instruments are calibrated and have valid calibration certificates.

8. Commissioning Strategy

As part of preparation for commissioning of the gas gathering pipelines and the field manifolds infrastructure, there is need to confirm all gas wells have been tested and producing with all parameters within the operating envelope.

On the BBOU manifold and ELEM Manifold, all valves on the manifolds should be serviced and tested before commencement of commissioning.

The pipeline from BBOU to ELEM manifold to be pressure tested and test result sheets confirmed.

The pipeline from ELEM manifold to the Slug catcher to be pressure tested and test result sheets confirmed.

8.1. Steps for Commissioning and Start-up:

- Pipeline Sub-system Testing (End Facilities)
- Pipeline Commissioning
- Final Dewatering Operations using air
- Displacement of trapped air using inert gas i.e., Nitrogen.

Sequence of events:

- a. Conduct Start-Up on paper
- b. Conduct Pre-Startup Audit and Safety Review.
- c. Obtain Ready for Start-up (RFSU) Certificate.
- d. Conduct Pipeline control system testing
- e. Conduct Pipeline commissioning

9. RFSU/PSSR/PSUA

Before the commencement of commissioning activities, there is need to carry out a review to ascertain that all works have been completed in line with the design, and also the Operations organization to operate the facilities are available and trained.

RFSU is considered as the point when all activities necessary to support the introduction of hydrocarbons, including all utility and process utility, safeguarding and shutdown systems have been pre-commissioned, commissioned and integrity verified.

9.1. PSUA / PSSR

A Pre-Start Up Audit (PSUA) shall be carried out prior to the initial start-up of an asset. The PSUA is a formal process carried out by an approved external or internally agreed project audit team and is used to determine the following;

- Facilities are constructed as per design;
- Operations philosophy is complied with;
- Asset integrity is verified and can be demonstrated;
- Necessary vendor support identified and scheduled;
- Commissioning 'imperatives' are in place and complied with;
- Project assurance in place ;
- HSE-MS is in place.

It may be preferable for the project to carry out a Pre-Start up Safety Review (PSSR) prior to the PSUA to provide the project with the assurance that all preparations for readiness for start-up are in place and are being managed effectively. This is to ensure all necessary pre-conditions for Facility start-up are in place. The PSSR shall aim to identify actions that need to be completed prior to the formal request for a PSUA.

10. Project Hand-Over

Efficient and effective Commissioning program requires the use of a variety of technical documentation. This documentation, which is specified below, must be on hand or readily available prior to initiating pre-commissioning activities.

10.1. Required Documents

All up to date Process Flow, Process Engineering and Safeguarding systems.

- Operating Manuals
- Design Books.
- Operator Standards.
- Construction Company Engineering Standards.
- All Relevant Codes.
- Chemical Cleaning Procedures which may be applicable if there are signs of internal rust in the pipelines.
- Flushing, Blowing and Drying Procedures.
- Instrument and Electrical Data Requisition Sheets.
- Instrument and Electrical Diagrams.
- Functional Testing Procedures.

The following items are all to be stored on site and are filed under the respective equipment number or tag number.

10.2. System Handover Documentation

Mechanical Completion and Pre-commissioning mean fabrication, construction, physical installation, testing, inspection, and system verification is complete, such that the Facilities are ready for introduction of hydrocarbon.

After the Construction Company has remedied these punch list items, with the exception of items agreed by the Commissioning Manager the system or sub system is deemed to be mechanically complete, and a Mechanical Completion Certificate will be signed by the required parties.

Any exceptions indicated on the Mechanical Completion Certificate such as insulation, painting or missing parts of minor importance have been confirmed by the Commissioning Manager that will not interfere with pre-commissioning activities or delay the start-up.

On completion of all pre-commissioning activities a comprehensive list of documents is required to hand-over the completed priority system as 'Ready for Start-Up' to the Operations team. The list should include: -

- Hydro tests acceptance.
- Welding/ x-ray reports.
- PSV test certificates.
- Vessel inspection and testing reports.
- Air blowing/Flushing procedures (Marked up P&ID's)
- Motor test runs.
- Instrument loop check

10.3. Operations P&ID Check List

The following check list is a guide of the minimum checks to be carried out on the latest P&ID drawing and assist the Operations Team in systematically checking the accuracy of the document.

Each Operations Team must use their plant knowledge and competences to ensure the check is carried out fully and completely, as vital information may be omitted if the line walk, and checks are not stringently followed, and any suspected deviations recorded.

It is important that all items on the check list are clearly defined, and the item number entered on the P&ID, and anomalies identified clearly for further investigation.

When unsure of the List details, validity of any anomaly, plant detail or layout issue, should be reported to the Project Engineer for further clarification.